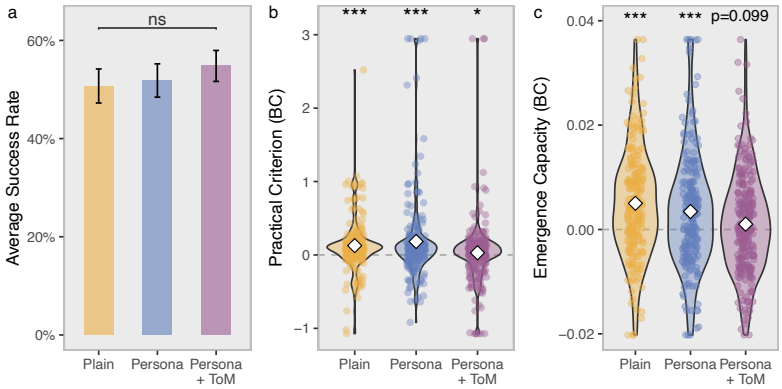

Figure 2: **a)** Group success across three interventions. **b)** Practical emergence criterion (bias corrected). **c)** Emergence capacity dynamical synergy (bias corrected). Data Winsorized at the 1st and 99th percentiles for visual clarity. Stars indicate significance level of Wilcoxon test. Notes: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

data. The bias-corrected estimates are well above 0 in all conditions (**Plain**: $p = 1.5 \times 10^{-16}$; **Persona**: $p = 6.6 \times 10^{-7}$; **ToM**: $p = 0.02$). We report additional robustness tests using (a) raw data, (b) functional null model residuals, (c) using agent reactivity as data, and (d) bias correction using Jeffrey's smoothing and Miller-Madow in the Appendix.

We repeat analyses using the emergence capacity criterion. We find it is significantly above its null across conditions (Figure 2c); results hold under residualization and alternative entropy corrections (see Appendix for details). Taken together, across two different tests (one using a macro signal the other using future system behavior) and across a variety of robustness tests we find evidence of dynamic emergence capacity in multi-agent LLM systems (answering **RQ1**).

4.2 Dynamical Mechanisms

How is this emergence dynamically maintained and where is it located within the groups? We find time-trend demeaned bias-corrected I_3 around 0 in the **Plain** (Wilcoxon p-value 0.974) and **Persona** condition ($p = 0.846$) indicating groups that fail to escape chaotic, uncoordinated states. In contrast, the **ToM** shows significant positive mutual information ($p = 3.5 \times 10^{-14}$), collapsing the group's behavior into a predictable macro-state (Figure A2). The number of groups with significantly positive I_3 is substantially higher in the **ToM** condition (Figure 3a), consistent with enhanced collective intelligence. To formalize this transition dynamic, we compute the Total Stability of the system (computed as I_3 normalized by the entropy of the macro-signal) which serves as a proxy for the Lyapunov stability of the collective state (Haddad & Chellaboina, 2008). We find Total Stability indistinguishable form zero in the **Plain** (Wilcoxon p-value 0.976) and **Persona** (p-value = 0.858) indicating a "gaseous" state where groups fail to settle into a stable attractor. However, the **ToM** intervention induces a sharp increase in Total Stability (p-value = 2.9×10^{-14}). Theoretically, this suggests the **ToM** prompt acts as a control parameter, steering the multi-agent system from a chaotic regime into a deep basin of attraction where collective behavior stabilizes and information processing capacity spikes: turning the system from a collection of individuals into a collective group.

Decomposing this stability reveals the specific mechanism of coordination. While the system exhibits emergence capacity and practical emergence, we find the stability in group behavior is supported by pairwise alignment rather than irreducible triplet complexity. Specifically, triadic information gain (G_3) is around 0 in **Persona** and **ToM** (Figure 3c). The **Plain** conditions shows small positive $G_3 > 0$ (Wilcoxon p-value = 0.026) but given the near-zero Total Stability, this likely reflects transient stochastic correlations and oscillation rather than sustained coordination. In dynamical systems terms, the **ToM** prompt creates a deep basin of attraction where agents converge on a shared schematic (or Schelling point), making the collective state robust but informationally redundant. This mirrors "Mean Field" coupling in physics: because agents receive only global feedback and cannot observe individual peers, they couple to the aggregate signal rather than forming

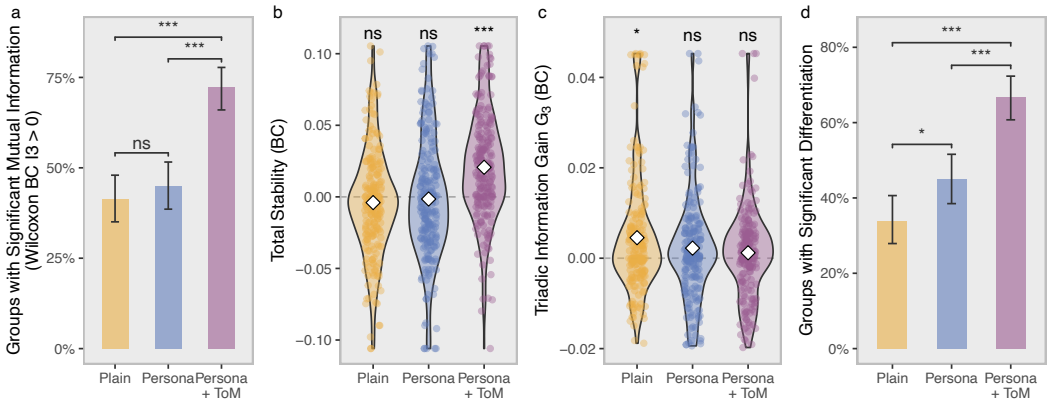

Figure 3: **a)** ToM-prompt condition has substantially more groups with significant I_3 content (above 0). **b)** Total Stability (time-delayed mutual information of triplets (I_3 normalized by macro-signal entropy, bias corrected). **c)** Information gain of triads over most informative dyad (G_3, bias corrected). **d)** Agent differentiation using hierarchical mixed model comparison (counting groups in which at least one test (different intercepts or slopes) is below $p < 0.05$. In panel a) and d), error bars show Wilson confidence intervals for binary data. Stars indicate significance level of test for equal proportion. Panel b) and c) shows bias corrected data with Jeffreys' prior. Data are Winsorized at the 1st and 99th percentiles for visual clarity. Stars indicate significance level of Wilcoxon test. Notes: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

distinct local bonds. In such an environment, attempting complex higher-order synergy would be fragile; instead, the system converges on the efficient solution of dense pairwise alignment to the Mean Field.

Next, we explore whether agents evolve specialized roles and identities. Are there detectable between-agent differences in either the level of their contribution to the group guess or their temporal evolution (i.e., learning rate)? Using the hierarchical model comparison test we find that agents in many groups have differentiated identities (Figure 3a; a Fisher test for joint p-value evidence is again highly significant). There is substantially more differentiation among agents in the **Persona** condition, and even more in the **ToM** condition. The reasoning traces of agents often contain references to the "personal experience" of their assigned persona such as "In my experience (whether it's corraling cattle or wrangling numbers), starting toward the middle gives the group the best shot to zero in after the first feedback" (see Appendix for additional sample quotes). In Appendix A.10 we provide additional robustness tests ruling out heterogeneous learning rates as the sole source of differentiation and provide additional evidence for stable agent identities that persist over time. That is, in the **Plain** condition, differentiation results only from idiosyncratic noise and transient drift introduced by probabilistic LLM responses. With **Persona**, agents gain stable, identity linked behavioral preferences that refine their differentiation. The addition, **ToM** sharpens role differentiation through feedback loops from mutual predictive modeling by conditioning on the public history (which functions as a common ground and coordination device; Lewis, 1969; Skyrms, 2010). This converts small persona-induced asymmetries into stable, self-reinforcing roles, creating the basis for strategic complementarity and stable alignment (answering **RQ2** and **RQ3**)

4.3 Functional Role of Emergence in Group Performance

What functional differences—if any—does emergence allow? We use regression analysis to explore the joint effect of synergy and redundancy (based on emergence capacity formulation) while carefully controlling for endogeneity of variables measuring emergent properties and performance and controlling for sample selection bias with stabilized inverse probability weighting (see Appendix A.11 for method details). On their own, higher levels of either synergy or redundancy do not predict success. However, when both are present, performance improves significantly (significant interaction with $\beta = 0.24$; $p = 0.014$). In marginal-effect terms, redundancy amplifies the benefit of synergy on the log-odds scale by 27%; and vice versa synergy amplifies benefits of redundancy

by 27%. This pattern implies that systems benefit when redundant pathways create goal alignment, while synergistic interactions extract novel, non-overlapping information—together enabling higher overall performance. We complement the regression analysis with causal mediation analysis (Imai et al., 2010). While reaching only marginal significant levels the effect is consistent: the **ToM** treatment causally increases performance indirectly by increasing synergy (ACME $= 0.034$ [95%CI: $-0.000 - 0.07$], $p = 0.053$). This aligns with the interpretation that performance benefits emerge when systems achieve both redundancy (aligned toward a common goal) and synergistic integration (differentiated, complementary roles)—a form of functional and collective complexity (Varley et al., 2023; Luppi et al., 2024; Sterelny, 2007).

4.4 Experiments using Other Models

To assess generalizability of our findings, we repeat experiments with four other models: Llama 3.1 8B and 70B, Gemini 2.0 flash, and the reasoning model Qwen3 (see Appendix A.13 for full details). We find that the capacity for emergent synergy is robust across high-capability models: Llama 70B, Gemini, and Qwen3 achieved success rates on par with GPT 4.1 and exhibited strong evidence of emergence using both the emergence capacity and practical criteria. Consistent with our main results, these high-capability models achieved higher success rates, more differentiation, and stronger evidence of emergence in the **Persona** and **ToM** conditions compared to the **Plain** baseline. Conversely, the smaller Llama 8B largely failed to break oscillatory cycles and develop goal-directed cross-agent synergy due to its lower ToM reasoning reasoning capacity (Xiao et al., 2025). Crucially, our analysis of the reasoning model Qwen3 uncovered a distinct structural challenge we term *paralysis under coordination ambiguity*. Qwen3 agents enter infinite chain-of-thought loops when attempting to reconcile local binary search strategies with noisy group feedback. We provide a detailed taxonomy of these reasoning failure modes—including mutual mental modeling traps—in Appendix A.13, highlighting a critical frontier for research of reasoning models in multi-agent systems (Piedrahita et al., 2025).

5 Related Work

Shortly after the release of ChatGPT, the community began exploring multi-agent LLM systems, where multiple LLM agents interact with each other. One early example is Park et al. (2023), who simulated a small population of "individuals" in a *The Sims*-style environment with friends, houses, and jobs, showing the emergence of rich social behaviors such as coordinating a Valentine's Day party. This was followed by multi-agent systems for complex tasks such as software development (Chen et al., 2023; Hong et al., 2024; Qian et al., 2023), healthcare (Li et al., 2024a), and other domains (Subramaniam et al., 2025; Ashery et al., 2025). Federated systems or "society of models" also gained attention (Juneja et al., 2024; Li et al., 2024b). These systems often achieve significant performance gains over single-agent baselines (Wu et al., 2023; Chen et al., 2023; Li et al., 2024b; Tao et al., 2024).

A recurring explanation for such gains invokes "greater-than-the-sum-of-its-parts" effects (Chen et al., 2023, p.1), often attributed to division of labor among differentiated agents (Subramaniam et al., 2025). Yet while this work is "inspired by human group dynamics" (Chen et al., 2023, p.1), it rarely evaluates whether the same principles that underlie effective (and ineffective) human groups also emerge in multi-agent systems. Two gaps stand out. First, human groups often coordinate by developing *role specialization* (Goldstone et al., 2024). Multi-agent systems often intuitively induce role specialization (such as "programmer", "tester", and "CEO"; Qian et al., 2023) yet have not systematically evaluated the impact on emergent shared cognition beyond win-rate comparisons. Second, groups only outperform individuals if their members specialize in contributing different pieces of information to the group task (Theiner, 2013; DeChurch & Mesmer-Magnus, 2010; Fazelpour & De-Arteaga, 2022; Nickerson & Zenger, 2004). Effective groups balance complementarity—members contributing distinct information—with redundancy, the alignment on shared goals (Theiner, 2013; DeChurch & Mesmer-Magnus, 2010; Fazelpour & De-Arteaga, 2022; Luppi et al., 2024; Tollefsen et al., 2013). Too much of either undermines performance. Unlike win-rate–centric evaluations—and tests of whether agents in multi-agent systems can sustain cooperation per se (Piedrahita et al., 2025; Piatti et al., 2024)—we propose falsification tests and null modeling, thus extending multi-agent frameworks with formal theory. While these principles likely apply to